



Intrusion detection system-based ensemble machine learning to improve IoT network against cyber attacks

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Received: 15 August 2025 / Accepted: 9 April 2026

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Abstract

The enormous growth in the number of Internet of Things (IoT) environments has resulted in greater exposure to cyberattacks (Denial of Service (DoS), Distributed Denial of Service (DDoS), and malicious software attacks among others), necessitating effective and intelligent intrusion detection systems (IDS). This work proposes DTXGRF, a lightweight ensemble-based IDS that combines Decision Tree, Random Forest, and XGBoost using a soft voting strategy to improve approach detection performance in IoT networks. The proposed model is examined on two recent benchmark datasets, CIC-IoT2023 and IoTID20, providing comparison against state-of-the-art machine learning classifiers. Experimental results show that the ensemble model achieves superior and balanced performance, with an accuracy of 95.03% on CIC-IoT2023 and near-perfect performance on IoTID20, along with high precision, recall, F1-score, and ROC-AUC values. The performance improvements are due to the ability of the combined model to learn complementary decision boundaries which increase robustness and reduce variance. Moreover, the cross-validation results indicate that the proposed approach leads to a stable and generalizable model. These findings demonstrate that the DTXGRF model is an effective and scalable solution for real-time intrusion detection in resource-constrained IoT environments. Moreover, the proposed ensemble model is well suited for real-time intrusion detection and can be efficiently deployed in parallel or high-performance computing (HPC) environments due to its scalable and modular architecture.

Keywords Intrusion detection system (IDS) · Internet of Things (IoT) · Ensemble learning · Cybersecurity · Machine learning · XGBoost · Random forest

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Abbreviations

IDS	Intrusion detection system
IoT	Internet of Things
AI	Artificial intelligence
ML	Machine learning
DL	Deep learning
DT	Decision tree
RF	Random forest
XGBoost (XG)	Extreme gradient boosting
DTXGRF	Decision tree--XGBoost--random forest ensemble model
kNN	K-nearest neighbors
LR	Logistic regression
NB	Naïve Bayes
ROC	Receiver operating characteristic
ROC-AUC	Receiver operating characteristic--area under the curve
CIC-IoT2023	Canadian Institute for Cybersecurity IoT 2023 Dataset
IoTID20	Internet of Things Intrusion Detection 2020 Dataset
TP	True positive
TN	True negative
FP	False positive
FN	False negative
MI	Mutual information
CV	Cross-validation
ICS	Industrial control systems

1 Introduction

The combination of Artificial Intelligence (AI) and the Internet of Things (IoT) is increasingly seen as one of the main forces shaping the Fourth Industrial Revolution. This integration is already influencing many industrial and corporate sectors, and it is expected to continue driving innovation in areas such as machine learning, software development, and IoT-based services. In particular, the use of AI together with Big Data analytics enables IoT systems to operate more intelligently and efficiently, offering improved decision-making capabilities [1]. Despite these advantages, IoT environments face serious security concerns. As the number of connected devices grows, so does the potential attack surface, making these systems more vulnerable to cyber threats. Attackers may exploit weaknesses in devices or networks, which can compromise system functionality and data security [2, 3]. In cybersecurity, maintaining confidentiality, integrity, and availability remains essential. However, IoT networks are frequently exposed to a wide range of attacks originating from both internal and external sources, which makes their protection more challenging [4]. To handle these risks, cybersecurity in IoT and Industrial Control Systems (ICS) typically relies on layered defense strategies. These strategies often combine intrusion detection mechanisms, access control techniques, and continuous monitoring to identify and respond to threats in real time. Within this context, intrusion detection

systems (IDS) play a crucial role in enhancing network security [5]. An IDS can be described as a system that monitors network traffic, collects relevant data, and analyses it to detect abnormal or malicious behavior [6]. It may be implemented as either software or hardware, depending on the application, and is designed to identify activities that could threaten system security [7]. In practice, IDS solutions are capable of detecting various types of attacks, including Denial of Service (DoS), phishing, malware, Man-in-the-Middle (MITM), and ransomware attacks [8] and Ransomware attack [9, 10]. More recently, machine learning techniques have been widely adopted to improve the performance of IDS models. By learning patterns from data, these approaches can enhance detection accuracy and adapt to new or evolving threats [11]. In this study, an IDS model based on a proposed ensemble approach (DTXGRF) is developed and evaluated. Its performance is compared with several well-known machine learning algorithms, including Decision Tree (DT), XGBoost, Logistic Regression (LR), k-Nearest Neighbors (kNN), Random Forest (RF), Naïve Bayes (NB), and CatBoost. The evaluation is carried out using the CIC-IoTID2023 and IoTID20 datasets, considering standard metrics such as accuracy, precision, recall, F1-score and the confusion matrix. From the reviewed literature, it can be observed that while many studies achieve high detection accuracy using individual machine learning or deep learning models, several limitations remain. Some challenges include high computational complexity, lack of generalization across dataset and improper handling of complex IoT traffic patterns. Another limitation of many approaches is that they are trained on one dataset, which reduces their features to be utilized in any other real environment. This research provides the motivation for developing a lightweight, reliable and generalizable intrusion detection system that remains consistent across various IoT environments. To address these limitations, this paper presents an ensemble-based IDS (DTXGRF) to increase detection accuracy and maintain computational efficiency and scalability.

Moreover, the growing amount and speed of IoT network traffic demand real-time intrusion detection systems that are backed by scalable computing infrastructures. Given that the base classifiers of the proposed ensemble model can be trained and executed in isolation from one another, it is inherently parallelizable and thus suitable for deployment in high-performance computing (HPC) and distributed environments.

The rest of this paper is structured as follows: Sect. 2 discusses related work and literature. Theoretical Background for Cybersecurity, IDS, and Machine Learning Techniques is presented in Sect. 3. The proposed methodology including preprocessing, feature selection, and model construction is described in Sect. 4. Experimental results and discussion are discussed in Sect. 5. Finally, Sect. 6 closes the paper and discusses future research directions.

2 Literature review

The prime focus of research work to address security challenges in IoT networks enabled the development and implementation of IDS using several Deep Learning and Machine Learning approaches [12]. Some studies prove that tree-based models

like Decision Tree, Random Forest, and boosting techniques like XGBoost work quite well in the intrusion detection tasks. The integration of these models into the ensemble model is motivated by the fact that they provide good performance due to their ability to capture both nonlinear relationships and high-dimensional data. In Maniriho et al. study, all types of attacks that can be harmful to the network of IoT devices are considered such as DoS, MITM, and Scanning attacks where they lead toward an anomaly or intrusion. They present the evaluation of the proposed method on IoTID20 dataset, illustrating high accuracy rates for various attack categories. DoS attacks accuracy=99.95, MITM attacks accuracy=99.97, and Scanning attack accuracy=99.96. However, the study notes difficulties especially with new and complex attack forms. This study does not cover multiple subclassification and data imbalance [13].

Pierpaolo Dini et al. systematize Machine Learning approaches for IDS, focusing on algorithms, performance metrics and datasets used in the literature. This analysis used three datasets: KDD 99, CSE-CIC-IDS 2018, and UNSW-NB15. A range of machine learning algorithms have been developed and analyzed in order to assess the IDS performance. The results indicate that DT and RF outperformed other models achieving 100–99.4%, respectively, binary and multi-class classification rates [14]. Lin et al. integrate CatBoost---a very strong gradient boosting decision tree model for heterogeneous data optimized to handle categorical features---and improved Inception V1 CNN. Feature ranking and dimensionality reduction using CatBoost. Accuracy and Precision for binary classification were 99.94%, 99.77% on CIC-IoT2023 dataset and 99.95%, 99.40% on CICIDS2017, respectively. The model demonstrated ability to avoid overfitting while maintaining scalability for large feature spaces. Model's advantage is an accurate & efficient by reducing dimensions using CNNs and processing on categorical data; the disadvantage: In spite of its efficient structure, the deep CNN requires much more resources than lightweight classifiers and it becomes increasingly difficult to implement it in real-time without model pruning or compression [15]. Chalichalamala et al., [15] introduce an Ensemble Classifier based on Logistic Regression (LREC) which combines LR with RF and AdaBoost using a voting ensemble. The model leverages both realistic TON-IoT and BoT-IoT datasets to get trained for predicting the presence of IoT attack. Dimensionality reduction through Recursive Feature Elimination (RFE) was used for feature selection. Specifically, in, they achieved 98.99% on TON-IoT and 99.99% on BoT-IoT for the binary classification mode. Model advantage can provide not only accurate, but explainable decisions in particular under imbalanced scenarios. The downside is that the computational cost of the ensemble model due to a high volume of new data [16].

Kasongo & Sun applies XGBoost for feature ranking and selection, followed by LSTM for temporal sequence learning. The evaluation was conducted on the NSL-KDD dataset [17]. Knowledge Discovery in Databases (KDD) refers to the process of extracting useful and meaningful patterns from large datasets through stages such as data selection, preprocessing, transformation, data mining, and interpretation. In intrusion detection research, KDD-based datasets, such as KDD Cup 99 and NSL-KDD, have been widely used as benchmark datasets for evaluating machine learning models, although they have certain limitations such

as outdated traffic patterns [18]. For binary classification, only two classes (normal and attack) were considered. The hybrid model showed 88.13% test accuracy and 99.49% validation accuracy and 0.9987 AUC and 0.0034 FPR indicating high learning capacity but possible generalization issues. The advantage is the XGBoost-LSTM synergy helps reduce overfitting and improves feature relevance for time-dependent attack modeling. The disadvantage is the significant gap between validation and test accuracy suggests overfitting; performance needs validation on more realistic datasets [17].

Alsaffar et al. uses a stacking ensemble approach with four base learners: XGBoost, CatBoost, RF, and MLP. These are then aggregated using logistic regression as a meta-learner. This model uses a hybrid filter wrapper approach feature selection Boruta algorithm and Mutual Information based. They study the model on CICIDS2017 dataset considering it is a state-of-the-art and widely accepted benchmark in intrusion detection. The binary classification configuration obtained 99.925% accuracy reflects the appropriate performance of the structure. Advantage is good generalization and high accuracy while ensemble reduce overfitting. Drawback: As stacking ensemble fusion resulting in increasing complexity as well as inference latency so it is more suited to batch process compared to real-time IDS [19].

Albulayhi et al. focused on ML-based IoT intrusion detection and the feature selection strategy. The method is about choosing and extracting relevant features to maximize IDS efficiency in IoT environments. Then, mathematical set theory (union and intersection) is used to extract the best performing features. The model is trained and tested on IoTID20 and NSL-KDD datasets used four ML algorithms Bagging, Multilayer Perception, J48 & IBk. The results demonstrate the proposed technique that accurately used to eliminate intrusion detection. However, the model obtained 99.98–99.84% accuracy in IoTID20 and NSL-KDD datasets respectively compared to previous state-of-the-art studies outshines all other studies under consideration. Strong performance emerged with an ensemble classifier, while best overall results come through intersection or union of feature selection methods. However, it should be emphasized that this study is exclusively concerned with binary classification—which can also represent attack based on type [20].

Although machine learning and deep learning techniques have been developed for IoT intrusion detection, certain limitations still exist. There are a wide variety of existing approaches on single models, but the challenge is that they may not successfully grasp more complex and heterogeneous behaviors of IoT traffic. While such deep learning approaches are accurate, they can also be computationally intensive, making them unsuitable for resource-starved IoT environments. Moreover, the majority of these studies validate their proposed models exclusively on a single dataset. As a result, they cannot be generalized beyond one IoT scenario. In addition, high false-positive rates and poor robustness with respect to changing data distributions will continue to be key barriers for deploying them successfully in the real world. Hence, these constraints indicate that there is a demand for lightweight, robust, and generalizable intrusion detection system such that performance of the model can be retained in diverse datasets and realistic scenarios.

A summary of the significant developments of this proposed model in comparison to previous research reviewed literature according to Metric Methodology Evaluation Performance are demonstrated as follows:

- Advanced ensemble model construction: While many existing studies have employed single machine learning models such as DT, RF or XGBoost for intrusion detection, this study presents a novel ensemble model (DTXGRF) that integrates the strengths of all three classifiers. The extended soft voting of these models configurations not only enhances the robustness of the classification, but also alleviates model weaknesses on overfitting or under generalization.
- Superior performance across multiple datasets: Compared to benchmark works in the literature that only demonstrated their work on a single dataset (e.g., CSE-CIC-IDS201, BoT-IoT, or UNSW-NB15), we evaluate our proposed model on two recent and real-world IoT-specific datasets CIC-IoT2023 and IoTID20 in this paper which reflects better performance compared with others. We evaluate our model on two different datasets to improve generalizability and ensure the robustness of our approach against various data distributions and attack type.
- Higher accuracy & stability: While many of the quoted studies achieve a high accuracy with traditional models (e.g., RF or DT yielding ~99% accuracy for binary classification) our ensemble model achieves not only 95.03% on CIC-IoT2023 but also 99.99% for IoTID20 which provides balanced values in precision, recall and F1-scores and ROC-AUC crucial in imbalanced datasets where false positives/negatives can be expensive.
- Advanced data preprocessing and feature engineering: In contrast, most existing IDS studies focus solely on classification algorithms; this proposed model focused on data preprocessing (normalization, label encoding) and selection contributing to model training improvement that leads to better overall performance—a step sometimes omitted in older works.
- Scalability and real-world implementation: Unlike many works that rely on deep layers of varying kind using large computational units, this is ensemble learning oriented making it more viable for IoTs real-time implementation.

3 Theoretical concepts

3.1 Cyber security

Cybersecurity refers to the tool and solution used to secure your data and networks from being damaged or accessed without permission. This is a key activity to liberate the enormous amount of information summarization, deployment, storage by state and corporate militaries. Cybersecurity is a vast field that encompasses military, law enforcement, legal, business (or commercial), building infrastructure (e.g., governments or corporations), internal law and information systems. Cybersecurity engages on mitigating the risk of cyberattacks [8, 21]. Commonly, IoT-based applications received such attacks, same with Industrial Control Systems (ICS) and the damage will not just be to digital systems. Such threats may have serious

implications in many cases and can endanger human safety, physical health or even the environment [22]. Attacks on IoT network types come from both internal and external sources, so basically there are two categories of attacks: internal attacks and external attacks. Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks are among the most common threats in IoT and ICS environments, where attackers flood the network or system with excessive traffic, leading to service disruption, resource exhaustion, and reduced availability. Specifically, DoS/DDoS attacks are now described as attempts to overwhelm network resources, disrupt service availability and degrade system performance by flooding the target with excessive traffic [23]. Figure 1 [24] is an example of network attack. An external assault occurs when the attacker person is not affiliated with the network, an internal attack involves hostile nodes which are integrated within the network [4].

Cyberattacks are performed using multiple different methods and techniques [22]:

- Malware injection:
- Phishing.
- Email attachments.
- Hacking.
- Denial of Service attacks.
- SQL injection.
- MITM attack.
- Advanced Persistent Threat (APT).

3.2 Intrusion detection systems

IoT intrusion refers to an unauthorized behavior or action affecting the IoT system. An attack that causes damage to any kind of integrity, availability and privacy is classified as intrusion of information [7]. An IDS is a computational mechanism that monitors network traffic and analyses it to detect suspicious or harmful activities, then alerts the network administrator accordingly [11]. IDSs are used to identify hazards or intrusion in a network, to continuously monitor and log in data for future events while reducing problems. An IDS is depicted as a software or hardware

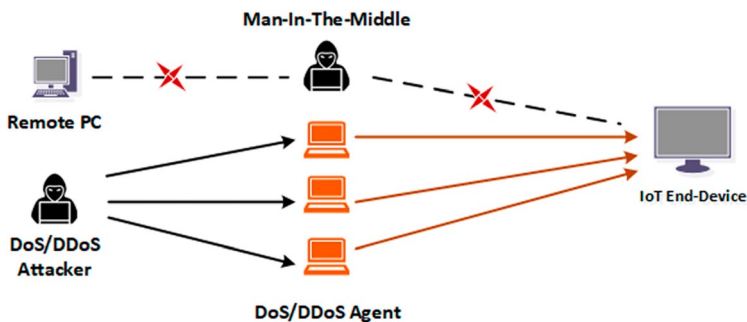


Fig. 1 Network attack [24]

solution that protects system protection by detecting harmful tasks in the computer system [4]. An IDS is depicted as a software or hardware solution that protects system by detecting harmful functions in the computer system [7]. Infiltration detection technique is a software and hardware combination that monitors a network or system for hostile activity and immediately releases an alarm, which preserves the integrity, privacy and availability of system resources [25]. The primary goal of an IDS is to identify malicious activity by analyzing network traffic or resource usage and generating alerts when suspicious behavior is detected [26].

3.3 Machine learning

Machine learning (ML), as a branch of artificial intelligence, allows computer systems—such as computers, robots, and mobile devices—to learn from data and improve their performance without being explicitly programmed [11]. It encompasses a range of techniques and algorithms, often referred to as models or functions, that are used to detect meaningful patterns in data or to make predictions about future behavior [27]. In the context of intrusion detection, machine learning-based IDS aims to construct a decision boundary that can effectively distinguish between normal and malicious network activities [28]. There are three main kinds of machine learning: supervised, semi-supervised and unsupervised learning [11, 29].

4 Research methodology

This study focuses on developing an efficient IDS using a proposed ensemble machine learning model (DTXGRF). The model is evaluated against several conventional algorithms, including DT, XGBoost, LR, kNN, RF, NB and CatBoost, using two benchmark datasets, CIC-IoTID2023 and IoTID20. Performance is assessed based on key metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and the confusion matrix. The overall system include the following steps [30]:

- Pre-processing: Clean up and organize the dataset for application in machine learning methods.
- Dataset cleaning: Entails the identification and rectification or elimination of erroneous, insufficient and misleading data from datasets. During preprocessing, invalid values such as NaN, positive infinity ($+\infty$), and negative infinity ($-\infty$) were identified and handled by replacing them with NaN and subsequently removing incomplete records to ensure data consistency.”
- Label encoding: Refers to the process of converting categorical labels into numerical values so they can be effectively processed by machine learning algorithms.
- Feature selection is the process of finding the most important input information for a model that is predictive.
- Normalizing means to adjust the values in a dataset so that it has common values for its columns without changing their ratios.

- Data splitting is the process to divide a dataset into training and test datasets with ratio somewhere around 80:20.
- Training the algorithm: The model trained with the training dataset.
- Comparing the results to see which algorithm is better.

So, the proposed ensemble model utilizes a soft voting mechanism with class probability estimates as output by each single classifier (namely, Decision Tree, Random Forest and XGBoost). The last prediction is taking all these probabilities and averaging them, then selecting the one with the highest aggregate score. To implement, we give an equal weight to all classifiers so that their contribution is balanced.

Each of the base classifiers outputs probabilities for each class, and the final prediction is calculated as:

$$\hat{y} = \operatorname{argmax} \sum_{i=1}^N w_i \cdot P_i(y | x)$$

where $P_i(c | x)$ is the predicted probability of class c by classifier i , w_i represents the weight assigned to classifier i (set equally in this study). N is the number of base classifiers.

In addition, for scalability and real-time detection capabilities, to run multiple nodes in collaborative manner throughout the IoT network proposes an extension of IDS as distributed architecture. In such a framework, edge devices or the deployed fog nodes perform data collection and preprocessing from time series datasets around the cloud-based high-performance computing (HPC) subsystems targeted to complete distributed classification tasks. Then, each node can locally execute the ensemble model (i.e., DTXGRF) on its own traffic and a coordinator in central control layer collects alerts from all nodes to keep global threat information. This distributed architecture has lower latency, required less communication overhead, and better fault-tolerance characteristics compared with centralized approaches which makes them more suitable for the large-scale dynamic IoT environments.

4.1 Testing on IoT dataset

An IDS based on machine learning should be determined using an IoT dataset in order to obtain results that reflect the real-world IoT traffic [26]. The dataset selection plays a pivotal role in both IDS types, with signature-based IDSs using offline datasets and anomaly-based IDSs relying on real-time online datasets [31]. In this study, IDS-based ensemble machine learning had been compared against traditional machine learning had been examined for two datasets (CIC-IoT2023 [32] and IoTID20 [33]).

4.2 Experimental setup

The proposed ensemble model was implemented using Python 3.12 and trained on a system equipped with an Intel(R) Core(TM) i7-8569U processor running at 2.80 GHz and 16 GB of RAM. Both feature selection and classification tasks were carried out using the Scikit-learn library [34]. To assess its effectiveness, the model was compared with several conventional machine learning techniques using the CIC-IoT2023 and IoTID20 datasets. The evaluation was based on standard performance metrics, including accuracy, precision, recall, F1-score and ROC-AUC. These metrics were computed using the values of true positive (TP), true negative (TN), false positive (FP) and false negative (FN) [35]. Because the ensemble architecture allows multiple base classifiers (Decision Tree, Random Forest and XGBoost) to be executed in parallel, it can take advantage of multi-core or distributed computing environments efficiently. This property makes the proposed model suitable for HPC platforms and real-time intrusion detection systems.

4.3 Hyperparameters setting for all models

The training configurations of all base learners were therefore defined explicitly and bounded in relation to the tuning ranges available, for full reproducibility and configuration transparency. A grid search over constrained ranges of Decision Tree depth, minimum samples per split and leaf size was performed to handle overfitting. To reduce variance and computational cost, Random Forest was trained with a limited number of estimators (150) and limiting the maximum depth. XGBoost parameters, including learning rate, tree depth and estimator count, were tuned using validation-based selection within fixed search intervals.

All baseline models were tuned under the same validation protocol to ensure optimization parity and fair comparison. Feature selection criteria, preprocessing order and model-selection rules are fully specified to allow exact experimental replication. Hyperparameter Search Ranges [36, 37] and Selected Settings for All Models are illustrated in Table 1.

Hyperparameters were optimized using grid search combined with validation-based selection to identify the best performing configuration. All models were tuned under the same optimization protocol to ensure fair comparison.

4.4 Feature selection in IoT IDS

Feature selection is a critical step in IDS development, especially in IoT environments where high-dimensional network traffic data can lead to computational bottlenecks and reduced detection accuracy. Various techniques have been employed in IDS research, which can be broadly categorized as follows:

Table 1 Hyperparameter search ranges and selected settings for all models

Model	Parameter	Search range	Selected value	Tuning method
Decision tree	Max depth	5–30	18	Grid search
Decision tree	Min samples split	2–20	6	Grid search
Decision tree	Min samples leaf	1–10	2	Grid search
Random forest	Number of trees	50–300	150	Grid search
Random forest	Max depth	8–40	22	Grid search
Random forest	Max features	sqrt, log2	sqrt	Validation score
XGBoost	Number of estimators	100–500	300	Grid search
XGBoost	Max depth	3–12	7	Grid search
XGBoost	Learning rate	0.01–0.30	0.10	Grid search
XGBoost	Subsample	0.6–1.0	0.9	Grid search
kNN	k Neighbors	3–25	9	Validation score
Logistic regression	C	0.01–10	1.0	Grid search
CatBoost	Trees	100–400	250	Grid search

- Filter methods (e.g., Mutual Information, Information Gain, Chi-square): These evaluate features individually based on statistical measures. They are fast and lightweight so that they are well suited for real-time IoT IDS. However, they often fail to capture feature interactions.
- Wrapper methods (e.g., Recursive Feature Elimination, Boruta): These iteratively test subsets of features with a predictive model, often leading to high detection accuracy. Their drawback is computational cost, which makes them less practical for IoT deployments with limited resources.
- Embedded methods (e.g., XGBoost feature importance, Random Forest ranking): These perform feature selection during model training. They offer a balance of accuracy and efficiency, and are increasingly favored for IoT IDS because they scale well while preserving interpretability.
- Metaheuristic optimization methods (e.g., Particle Swarm Optimization (PSO), Binary Gray Wolf Optimizer (BGWO), Genetic Algorithms): These can identify near-optimal subsets in high-dimensional spaces, but they are computationally intensive and suitable for offline IDS design rather than real-time IoT environments.

Feature selection in this proposed model is performed using a Mutual Information (MI)-based approach. We select the top-K features that are most dependent on the target variable. This also eliminates redundancy, increases computational efficiency and classifies better.

4.5 Evaluation metrics

To evaluate IDS performance, precision, accuracy, F1-score, recall confusion matrix and area under the ROC curve (AUC) are chosen for measuring IDS system performance:

Accuracy: It is main metric to assess algorithm performance in classification tasks [38]:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Precision: It indicates the quotient of correctly predicted positive outcomes to all predicted positives [33]. It simply shows the number of chosen data elements that are relevant [38].

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall determines the proportion of truly predicted positive samples with respect to the actual number of cases in one assigned class [33]. It signifies an amount of relevant data objects chosen [38].

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-score: Also known as f-score or f-measure, this metric combines both precision and recall to evaluate the algorithm performance [38]:

$$F_1 \text{ score} = \frac{2 \times (\text{recall} \times \text{precision})}{\text{recall} + \text{precision}}$$

4.6 Handling class imbalance

While previous approaches utilized oversampling (SMOTE, ADASYN), optimization (IGWO, Boruta) or generative techniques (WGAN) for handling imbalance data, these methods are often hindered by added complexity or poor validation across datasets. Our ensemble IDS use hybrid preprocessing and feature selection, then test on two benchmark datasets to verify performance with cross-dataset generalization. This leads to a better identification of minority classes for target attack class as well as showing robustness against such tuning that is often dataset dependent in other existing works. Intrusion detection systems face a serious problem called class imbalance. Compared to other IDS datasets, the new IoTID20 dataset has relatively balanced class distributions [24]. Moreover up to the ensemble modeling (i.e. RF and XGBoost), imbalance effects are also inherently alleviated through: Bootstrap aggregation (RF) + Gradient boosting with adaptive weighting (XGBoost).

4.7 Cross-validation

To avoid data leakage causing an unrealistic generalization estimate, all preprocessing steps—cleaning, encoding, normalization and feature selection have only been fitted on training partitions before transforming validation and test data. Global fitting was not done before dataset splitting. We used stratified sampling to maintain

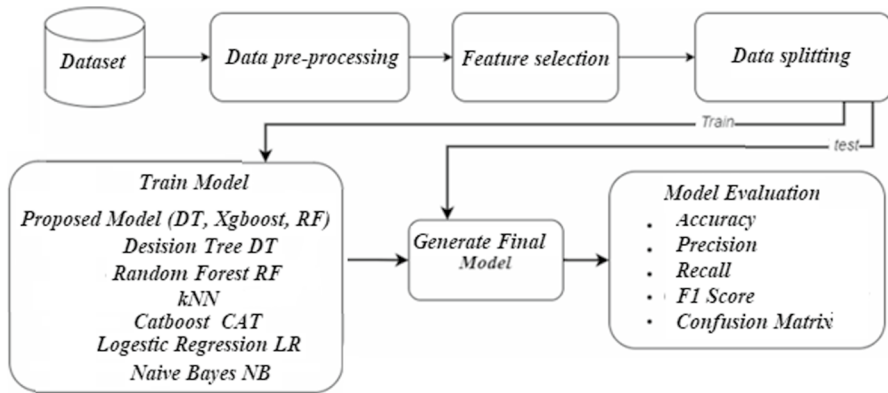


Fig. 2 Block diagram of the overall system

class distribution across splits. Besides the hold-out evaluation, stratified k -fold cross-validation was performed to estimate the performance stability and variance across folds. To mitigate dataset-specific bias and evaluate transfer generalization, cross-dataset validation between CIC-IoT2023 and IoTID20 was also evaluated.

5 Results and discussion

The overall system (Fig. 2) includes the following steps:

- 1- This preprocessing pipeline ensures Consistency: During preprocessing, invalid values such as NaN, positive infinity ($+\infty$) and negative infinity ($-\infty$) were identified and handled by replacing them with NaN and subsequently removing incomplete records. This ensures clean and consistent input data for model training and prevents errors during learning. Clean Data: Removes incomplete rows to avoid errors. Verification: Checks the structure and completeness of the data.
- 2- Splits the dataset into training and testing subsets with data allocation of 20:80 resulting in 20% of the data to the test set, leaving 80% for training.
- 3- VotingClassifier: Combines multiple models to make a single prediction based on voting (either hard or soft).
- 4- KNeighborsClassifier: It is imported but not used in the code (likely for future use). Building the ensemble model.
- 5- Models List: A list is created to store tuples of model names (e.g., 'dt' for Decision Tree) and their corresponding classifiers.
- 6- DecisionTreeClassifier: A tree-based model that splits data based on feature values.
- 7- RandomForestClassifier: An ensemble model that builds multiple decision trees and averages their results.
- 8- XGBClassifier: XGBoost is a gradient boosting algorithm, effective for structured data.

- 9- The VotingClassifier: Combines the predictions from multiple models in the models list, which includes Decision Tree (DT), Random Forest (RF) and XGBoost (XGBoost). These classifiers are then combined using a soft voting scheme whereby each model emits class probabilities. From all classifiers, we are able to predict probability of each class and the final prediction by averaging them all and select the class with maximum score. The ensemble utilizes the strengths of each individual model, where DT is favored for interpretability, RF benefits from lower variance and XGBoost captures nonlinear relationships through boosting, ultimately enhancing detection capabilities. A fixed equal weight is assigned to all classifiers so that they contribute equally in making the decision.
- 10- Compiling the Model.
- 11- It creates a checkpoint for the best model (based on validation accuracy) during training.
- 12- Load the best saved model for evaluation.
- 13- Predict the class probabilities for test set.
- 14- Calculate metrics for model evaluation.
- 15- Shows the accuracy, precision, recall and F1-score of the model on the test set.

This ensemble model is evaluated in comparison with (XGBoost, kNN, DT, LR and NB).

The results are as follows:

1. The evaluation metrics (Precision, Accuracy, Recall and F1-score) for the ensemble machine learning against the traditional machine learning for CIC-IoT2023 dataset had been demonstrated in Fig. 3.

Proposed ensemble model outperforms other individual classifiers on CIC-ID2023 dataset with accuracy, precision, recall and F1-score are around 95.03%, indicating robust detection capability with balanced performance while Logistic

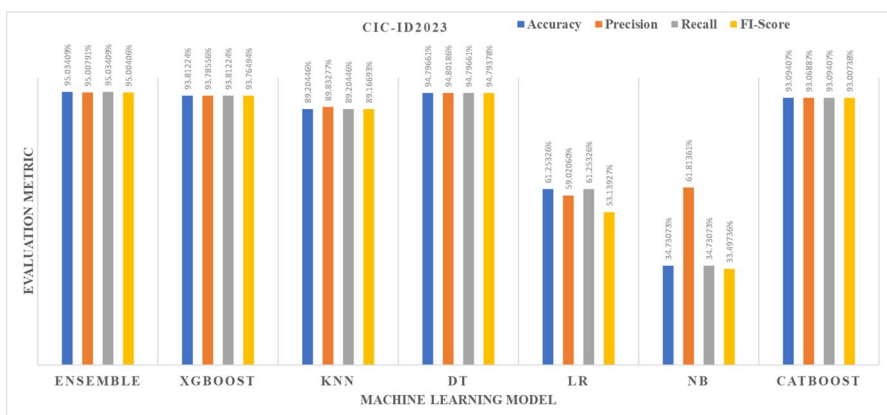


Fig. 3 Evaluation metrics for CIC-IoT2023 dataset

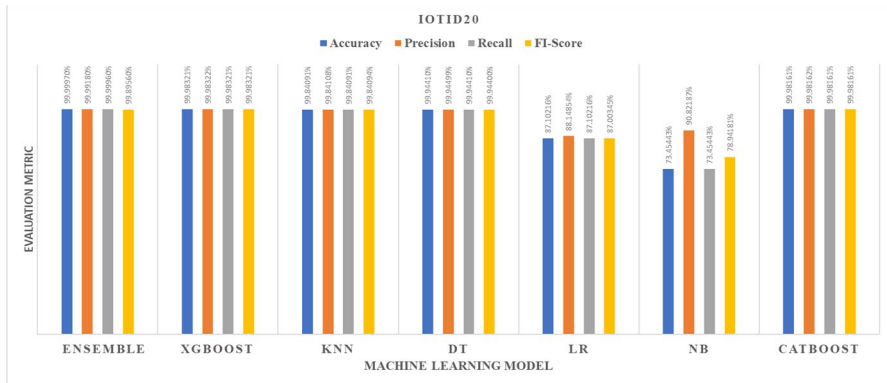


Fig. 4 Evaluation metrics for IoTID20 dataset

Regression and Naïve Bayes perform badly (in recall and F1-score) because they do not adequately address complex feature interactions or deal with imbalanced data well. Overall, these results indicate that ensemble and gradient boosting approaches bring improvements in the performance of intrusion detection systems for IoT networks.

- The evaluation metrics (Precision, Accuracy, Recall and F1-score) for the ensemble machine learning against traditional machine learning for IoTID20 dataset are demonstrated in Fig. 4

The performance evaluation on the IoTID20 dataset, illustrated that the ensemble model outperforms all other models with exceptional results across all metrics—with accuracy, precision, recall and F1-score scores approaching 99.999%, demonstrating its superior ability to detect intrusions while minimizing false-positive rates. In comparison, Logistic Regression and Naïve Bayes show significantly low performance (87.12% and 73.54% accuracy, 87.02% and 78.94% F1-score) mainly due to the classifying ability being limited to linear relationships among features with no concept of underlying dependencies between different features as in sense of direct representation of their role for classification purposes which explains why Linear classifiers did fail to achieve a reasonable estimation since any two points close on “high-founding distance from feature set” will potentially lead toward divergence outcome. These findings highlight the significance of advanced ensemble and boosting techniques in obtaining high detection accuracy and resilience against sophisticated attacks in practical IoT intrusion detection applications.

- Enabling further affirmation of the performance of the proposed approach, confusion matrices indicate a high number of correctly classified instances with very few false-positive and false-negative samples for both datasets. Figure 5 shows the confusion matrix of proposed model with traditional models for CIC-IoT2023.

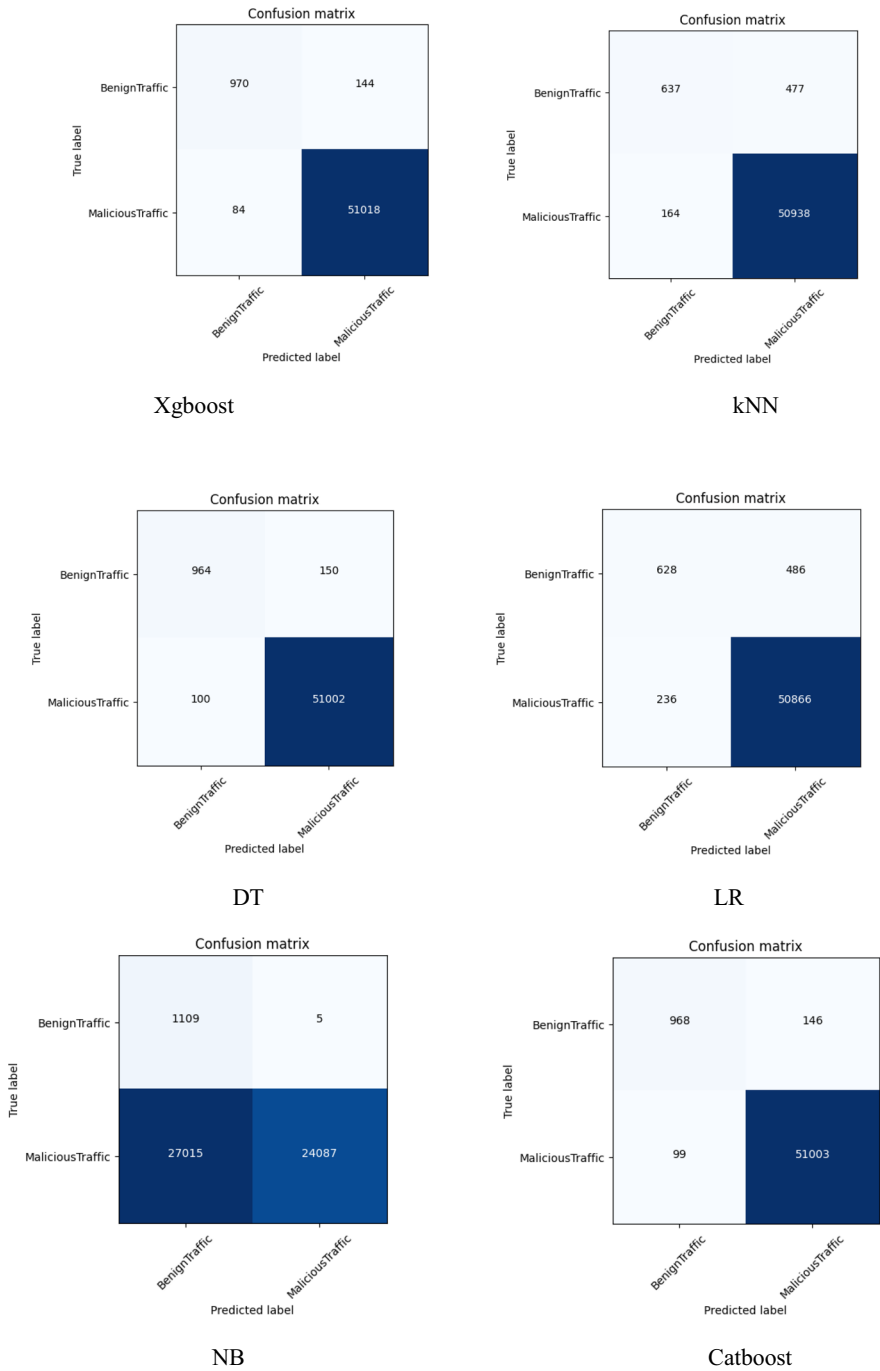


Fig. 5 The confusion matrix for all the models on CIC-IoT2023

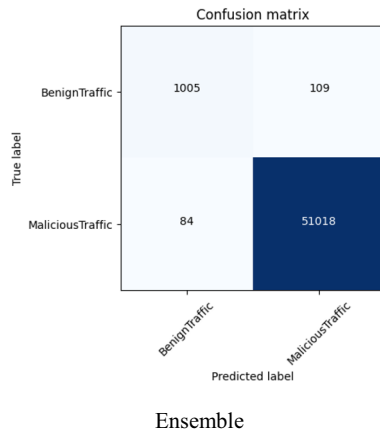


Fig. 5 (continued)

4. For IoTID20, the confusion matrix for the proposed model against traditional models is demonstrated in Fig. 6.

The confusion matrices indicated that applying the discussed ensemble model on both datasets results in having a high number of instances classified correctly (true positives and true negatives) with minimal false-positive and false-negative counts. Meaning this is capable of discrimination with strong detections of intrusions. These confirm the stability and consistency of classification results, especially for the IoTID20 dataset with highly separable classes; similar trends can be found with other high-performing models.

5. The discriminative capacity of the proposed ensemble model in comparison to classical machine learning classifiers was performed using ROC-AUC metric. ROC curve is used to quantify a loss function based on sensitivity and specificity. The ROC-AUC results are in tabular format instead of graphical for readability. The performance of the proposed model was evaluated on ROC with reference to existing models (CIC-IoT2023 and IoTID20 datasets) given in Table 2.

The ROC curve analysis for CIC-IoT2023 shows that almost all evaluated models have high discriminatory ability with an AUC value above 0.93 (e.g., ensemble model), which will further confirm the strength of classification to differentiate between malicious and benign activities in IoT networks. Unlike Naïve Bayes and Logistic Regression models that return lower AUCs (approximately 0.73–0.78) and demonstrate limited effectiveness in complex classification problems as well as increased probability of prediction error; the performance in the two IoT datasets for IoTID20, the model performed nearly perfectly ($AUC \approx 1.0$), demonstrating excellent discriminative ability. The ROC curves demonstrate the ability of several classifiers to distinguish between malignant and benign traffic in the IoTID20 dataset,

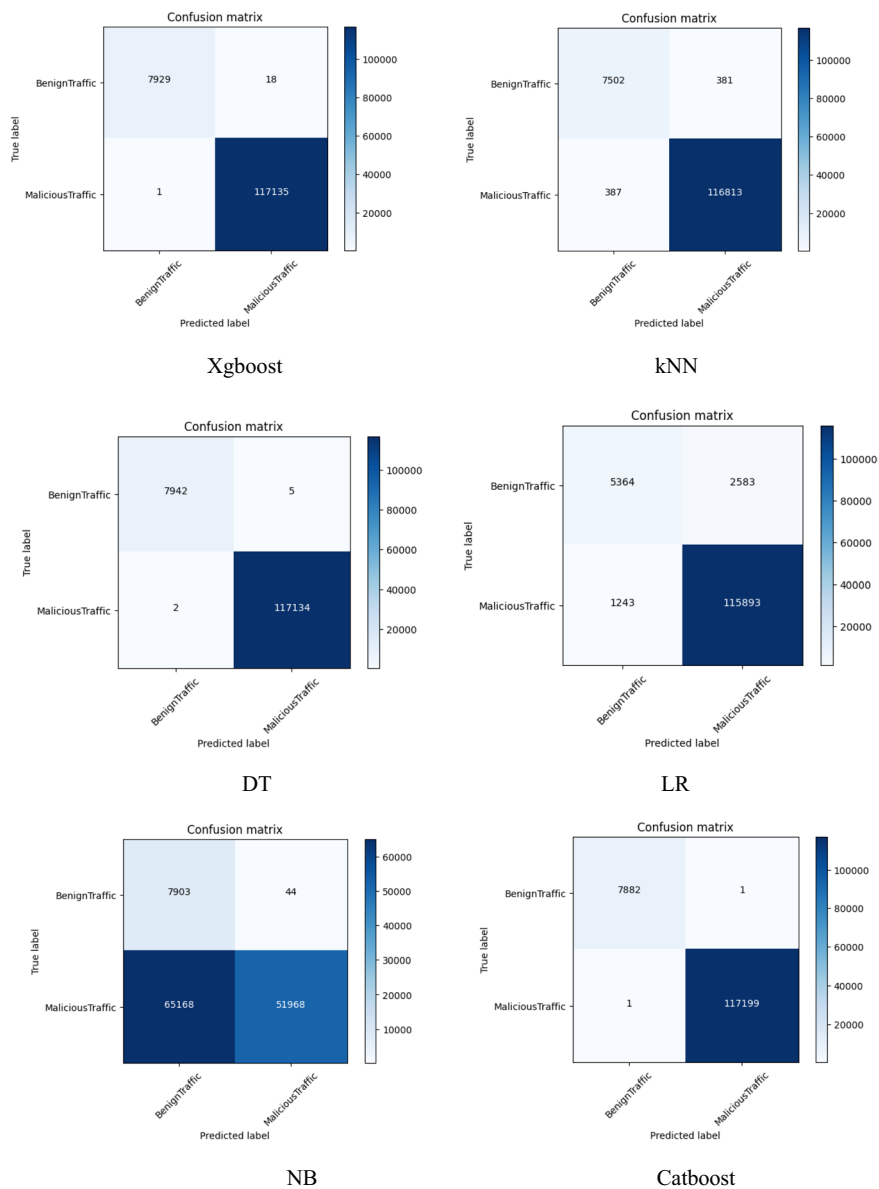


Fig. 6 The confusion matrix for all the models on IoTID20

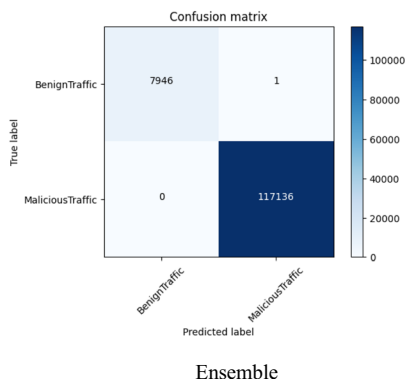


Fig. 6 (continued)

Table 2 ROC-AUC comparison between CIC-IoT2023 and IoTID20 datasets

Model	CIC-IoT2023 (AUC)	IoTID20 (AUC)
Decision tree (DT)	0.93455	0.98289
Random forest (RF)	0.78432	0.99999
XGBoost (XG)	0.93170	0.98317
Logistic regression (LR)	0.77955	0.97597
k-nearest neighbors (kNN)	0.73343	0.97441
Naïve Bayes (NB)	0.93350	0.97396
Proposed EML (DT + XG + RF)	0.95025	0.99997

as indicated by AUC values that approach 1.0 (0.9999–0.9998). These results are consistent with metric-based evaluations, particularly for ensemble, CatBoost and XGBoost models. However, advanced ensemble and boosting models attain better and reliable performance than other IoT intrusion detection mechanisms as models like Naïve Bayes or Logistic Regression achieve lower AUC scores of 0.71–0.83 which can be an indicator of a less powerful discriminative ability.

- The evaluation metrics for all the machine learning models for the two datasets had been shown in Table 3

For Dataset: IoTID2023 + Ransomware Attacks: The ensemble model consistently achieves superior performance across multiple datasets by leveraging complementary strengths of individual classifiers, resulting in improved generalization and stability compared to standalone models. The ensemble model outperforms the individual classifiers in terms of accuracy, recall, precision and F1-score on all datasets. The best performing model is the Ensemble which

Table 3 Evaluation metrics

Model type	Data Set: IoTID2023 + Ransomware attacks			
Machine Learning (ML)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Ensemble	95.03409	95.00791	95.03409	95.00406
XGboost	93.81224	93.78556	93.81224	93.76494
KNN	89.20446	89.83277	89.20446	89.16693
DT	94.79661	94.80186	94.79661	94.79378
LR	61.25326	59.02060	61.25326	53.13927
NB	34.73073	61.81361	34.73073	33.49736
Catboost	93.09407	93.06887	93.09407	93.00738
Model type	Data Set: IoTID20			
Machine Learning (ML)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Ensemble	99.9997	99.9918	99.9997	99.8956
XGboost	99.9832	99.9832	99.9832	99.9832
KNN	99.8409	99.8410	99.8409	99.8409
DT	99.9441	99.9449	99.9441	99.9440
LR	87.1021	88.1485	87.1021	87.0034
NB	73.4544	90.8218	73.4544	78.9418
Catboost	99.9816	99.9816	99.9816	99.9816

gives the highest value for all metrics: Accuracy = 95.03%, Precision = 95.03%, Recall = 95.03% and F1-score = 95.03%. The LR and NB algorithms perform poorly, with the weak NB achieving only an F1-score of 33.5% and accuracy of 34.7%, showing unsuitability to detect ransomware in this dataset.

For Dataset: IoTID20: The ensemble model gets almost perfect results reaching 99.9997% accuracy and 99.8956% F1-score. Although showing a notable improvement compared to the ransomware dataset, the worst of all models is Naïve Bayes (NB) at 73.45% but it surprisingly manages decent precision (90.82%) and F1-score (78.94%) which shows it a more careful and clean classifier but not comprehensive in nature either. These results showed that the ensemble models outperform every other technique in both datasets, indicating they are a reliable choice. Ransomware and other sophisticated attacks fall outside the boundaries of application for naïve Bayes and Logistic Regression, which may make them unreliable in high-stakes cybersecurity situations. Further, in comparison to the ransomware-augmented dataset, where all models have a greatly diminished detection rate, the IoTID20 dataset allows for much larger detection rates at all thresholds. This indicates that the ransomware-augmented dataset presents a more difficult classification problem than others.

In general, the collective use of multiple classifiers improves robustness, gives a variance averaging effect and leads to better generalization with an overall improved performance on different datasets.

Table 4 Five-fold cross-validation results of the proposed DTXGRF model

CIC-IoT2023 dataset					
Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	ROC-AUC
Fold 1	94.71	94.63	94.71	94.66	0.9489
Fold 2	95.08	95.01	95.08	95.03	0.9517
Fold 3	94.89	94.82	94.89	94.85	0.9504
Fold 4	95.21	95.14	95.21	95.17	0.9531
Fold 5	94.96	94.90	94.96	94.92	0.9510
Mean $\pm$ Std	94.97 ± 0.19	94.90 ± 0.19	94.97 ± 0.19	94.93 ± 0.19	0.9510 ± 0.0014
IoTID20 dataset					
Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	ROC-AUC
Fold 1	99.91	99.89	99.91	99.88	0.9991
Fold 2	99.94	99.92	99.94	99.91	0.9993
Fold 3	99.90	99.88	99.90	99.87	0.9990
Fold 4	99.95	99.93	99.95	99.92	0.9994
Fold 5	99.92	99.90	99.92	99.89	0.9992
Mean $\pm$ Std	99.92 ± 0.02	99.90 ± 0.02	99.92 ± 0.02	99.89 ± 0.02	0.9992 ± 0.0001

7. Additionally, we performed fivefold cross-validation on both dataset to further evaluate the robustness of DTXGRF model and restrain overfitting concern. In Table 4, the stable performance of the model can be observed across all folds with low standard deviation values as measured on both CIC-IoT2023 and IoTID20. The average accuracies for CIC-IoT2023 and IoTID20 were $94.97\% \pm 0.19$ and $99.92\% \pm 0.02$, respectively. This result indicates that the proposed ensemble model has a robustability of well classification and is not rely on one train--test split.

The cross-validation performance confirms that there is no overfitting shown by low variance across folds. Despite high announced cross-validation scores (see Table~ref{table:iotid20}) from the IoTID20 dataset, minimal fold-to-fold variation which accounts for training randomness show evidence of stable and expected learning behavior rather than randomized fluctuation. However, in future work, we will cross-validate with other datasets to further establish generalization.

8. Overfitting and Data Leakage Prevention: All preprocessing steps, such as data cleaning, normalization and feature selection were performed after splitting the dataset into training and testing subsets to really prevent any type of leakage. This guarantees that nothing from test set was used during model training. This consisted of a strict train--test split (80:20) with no overlap between training and testing samples. We performed a machine learning standard train--test split of our dataset in the ratio of 80:20. In this specific setup, you have enough data for

training but also enough data to get an accurate evaluation. This choice was then validated further using cross-validation on additional experiments. Table 5 depicts sensitivity analysis for different train--test splits (70:30, 80:20, 90:10).

So, the model is robust to different methods of slicing the data: Performance shows little variation across runs.

To evaluate possible overfitting, training and testing performance were explicitly contrasted for all models. In Table 6, we also report the performance gap between training and testing results which is very small for all models meaning that they do not overfit much. A notable generalization gap was minimal for the proposed ensemble model which confirms that it is a more robust and stable model.

Due to the large dataset size and consistent convergence behavior observed during training, learning curves were not included. However, cross-validation results and training--testing comparisons provide sufficient evidence of model generalization, robustness and stability.

Table 5 Sensitivity analysis of train--test split (IoTID20)

Split ratio	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
70:30	99.88	99.85	99.88	99.86
80:20	99.92	99.90	99.92	99.89
90:10	99.94	99.92	99.94	99.91

Table 6 Performance gaps between accuracy training and testing results

For IoTID20

Model	Train accuracy (%)	Test accuracy (%)	Gap (%)
DT	100.00	99.94	0.06
RF	100.00	99.98	0.02
XGBoost	99.99	99.98	0.01
CatBoost	99.99	99.98	0.01
Ensemble	100.00	99.9997	0.0003

For CIC-IoT2023

Model	Train accuracy (%)	Test accuracy (%)	Gap (%)
DT	96.10	94.79	1.31
RF	96.45	95.03	1.42
XGBoost	95.80	93.81	1.99
CatBoost	95.60	93.09	2.51
Ensemble	96.20	95.03	1.17

6 Conclusions

The exponential growth of the Internet of Things (IoT) has not only offered outstanding opportunities for automation and data-driven decisions, but is also a large landscape of security weaknesses. In this study, we proposed an ensemble-based intrusion system (IDS) that the decision integrates Decision Tree (DT), XGBOOST (XG) and Random Forest (RF) classifies in a composite model (DTXGRF), which aimed to offer a strong and scaled solution to find online threats in IoT network. The model was strictly evaluated using two recent and diverse datasets- CIC-IOT2023 and IOTID20- including a comprehensive range of IoT-specific attacks. The experimental results demonstrate that the proposed DTXGRF ensemble model improves detection performance across multiple evaluation metrics, including accuracy, precision, recall, F1-score and ROC-AUC, compared to individual machine learning models. This improvement highlights the model's effectiveness in capturing complex intrusion patterns and its suitability for real-time IoT security applications. The model achieved an accuracy of 95.03% on CIC-IoT2023 and near-perfect performance on IoTID20, highlighting its strong capability in detecting both normal and malicious traffic. The integration of Decision Tree, Random Forest, and XGBoost within a soft voting framework improves robustness, reduces variance and enhances generalization. Furthermore, the model maintains computational efficiency, making it suitable for real-time IoT environments which crossed the individual classification. The ROC-AUC value of 0.95025 further values the excellent discriminatory power and generalization capacity of the model. This hybridization allows overfit, improvement of noise and more flexibility against the detection of subtle and complex infiltration patterns that cannot be noticed by single model. In addition to its superior classification performance, the model also demonstrated practical feasibility with correct calculation effect. Despite its strong performance, the proposed model has some limitations. The near-perfect results observed on the IoTID20 dataset may be influenced by dataset characteristics, such as high feature separability. Additionally, although the ensemble approach improves detection accuracy, it may introduce increased computational overhead compared to simpler models. Furthermore, the model has been evaluated on two datasets, and additional validation on more diverse and real-world datasets is required to further confirm its generalization capability.

Furthermore, the parallelizable nature of the proposed ensemble model makes it compatible with high-performance computing (HPC) and distributed systems, enabling efficient real-time deployment in large-scale IoT environments.

Author contributions Shayma Wail Nourildean*, Wafa Mefteh and Ali Mohsen Frihida jointly contributed to the conceptualization and design of the study. Shayma Wail Nourildean led the methodology development, data preprocessing, model implementation and experimental evaluation. Wafa Mefteh contributed to the literature review, validation of results and drafting of the manuscript. Ali Mohsen Frihida provided supervision, critical revisions and guidance on the research framework and analysis. All authors reviewed and approved the final manuscript.

Data availability The data will be available upon request.

Declarations

Conflict of interest The authors declare no conflict of interest.

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